

Subject Description Form

Subject Code	ME42014
Subject Title	Networked Robots and IoT
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ME35002 Sensors and Actuators for Mechatronics
Objectives	The subject aims to equip students with knowledge of: 1. Fundamentals of Internet of things. 2. Networks of robots, sensors, and systems. 3. Hands-on project.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. mathematically model different networks that can represent multi-robot systems and IoT. b. integrate and process multimodal sensor data across complex networks. c. design algorithms to coordinate multiple robotic systems over a network. d. develop robotic/sensing systems to solve different practical problems. e. work in teams and conduct experiments related to networked robots and IoT technologies.
Subject Synopsis/ Indicative Syllabus	Introduction to IoT. Fundamental concepts; Application domains; Technologies and systems used in IoT. Integration of robotics with IoT. Sensing and control over networks. Modelling of networks. Fundamentals of linear algebra; Introduction to graph theory; Edges and vertices; Topologies; Undirected and directed networks; Laplacian matrices and network dynamics. Consensus protocols. Agreement in undirected networks; Agreement in directed networks; Agreement and Markov chains; Edge agreement; Stability of consensus. Collective decision-making. Cooperative networked robots. Formation control with shapes and relative states; Networks of mobile robots; Collective object transport; Task allocation; Coverage problem. Robot swarms. Sensor networks. Distributed sensing and estimation; Distributed linear least squares; Distributed Kalman filtering; Implementation over networks. Local communication. Practical work. Introduction of hardware used for IoT and networked robotics. Conduct a hands-on project where students will apply different concepts studied during the course.

Teaching/Learning Methodology	1. Lectures aim at providing students with fundamental knowledge required for understanding and analysing different networked systems, including its mathematical models, controller design, and algorithms. (Outcomes a to e) 2. Tutorials aim at enhancing students’ analytical and problem-solving skills in networked robots and IoT. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to e) 3. The experiments/project aim to provide hands-on experience for developing networked robots and IoT and reinforcing the acquired knowledge. (Outcomes a to e) <table><tr><th>Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th></th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Project or experiments</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes						a	b	c	d	e	1. Lectures	√	√	√	√	√	2. Tutorials		√	√	√	√	4. Project or experiments	√	√	√	√	√										
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Class test</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Coursework: Include assignments, projects, lab reports</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Final exam</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.40 x End of Subject Examination + 0.60 x Continuous Assessment 1. The continuous assessment including class test and coursework aims at evaluating the progress of the students’ study, assisting them in self-monitoring the respective learning outcomes, and applying the knowledge learnt in practical situations. 2. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	1. Class test	30%	√	√	√	√		2. Coursework: Include assignments, projects, lab reports	30%	√	√	√	√	√	3. Final exam	40%	√	√	√	√		Total	100 %					
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Reading List and References	• Lecture notes, articles, and handouts. • Graph Theoretic Methods in Multiagent Networks, by M. Mesbahi and M. Egerstedt. Latest edition. • Graph Theory You need Before Undergrad Research in 8 Undergrad Lectures, by M. Omar. Latest Edition. • Wireless Sensor Networks, by S. Chai, B. Zhang, et al. Latest Edition • Swarm Robotics: A Formal Approach, by H. Hamann, Latest Edition. • The Internet of Things, by S. Greengard. Latest Edition.
Develop Date	June 2024